LESSON 2.6

Divide Multi-Digit Numbers Using an Algorithm

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

WHY THIS MATTERS The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation. How many bars does each section receive?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Algorithm *Spanish: Algoritmo*	A step-by-step set of rules or instructions to solve a math problem.	Long Division Algorithm — D (Divide), M (Multiply), S (Subtract), B (Bring Down)
Dividend *Spanish: Dividendo*	The total number being divided into equal groups (the number inside the division bracket).	In 1,344 ÷ 12 = 112, the dividend is 1,344 — the total amount being divided
Divisor *Spanish: Divisor*	The number of equal groups you are dividing into (the number outside the division bracket).	In 1,344 ÷ 12 = 112, the divisor is 12 — the number of equal groups
Quotient *Spanish: Cociente*	The answer when you divide.	In 156 ÷ 12 = 13, the quotient is 13 — there are 13 groups
Remainder *Spanish: Residuo*	What is left over when a number does not divide evenly.	17 ÷ 5 = 3 R 2 means 3 groups of 5 with 2 left over
Divide Multi-Digit Numbers *Spanish: Dividir números de varias cifras*	Separate a multi-digit number into equal groups to find a quotient.	1,344 ÷ 12 = 112

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I want 1,344 ÷ 12. The dividend is 1,344 and the divisor is 12, so 12 goes outside the bracket and 1,344 goes inside.

1. DIVIDE: 12 does not fit into 1, so I look at 13. How many 12s fit into 13? One. I write 1 above the 3.
2. MULTIPLY: 1 × 12 = 12. I write 12 underneath the 13.
3. SUBTRACT: 13 − 12 = 1. That is what is left so far.
4. BRING DOWN: I bring the next digit, 4, down next to the 1 to make 14. Now I repeat the cycle.
5. REPEAT: 14 ÷ 12 → 1 (written above the 4), 1 × 12 = 12, 14 − 12 = 2, and I bring down the last 4 to make 24.
6. REPEAT: 24 ÷ 12 → 2 (written above the last 4), 2 × 12 = 24, 24 − 24 = 0. Nothing is left to bring down, so the remainder is 0.
7. The digits I wrote are 1, 1, 2, so 1,344 ÷ 12 = 112. I check by multiplying: 12 × 112 = 1,344.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's do 384 ÷ 8. DIVIDE: 8 does not fit into 3, so we look at 38. How many 8s fit into 38? (4 — write it above the 8.)

1. _____
2. _____
3. _____
4. _____
5. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- 2,184 ÷ 14 = _____ because I repeated divide, multiply, subtract, and bring down until there were no digits left to bring down, and what was left at the end was _____.

Word bank Algorithm | Dividend | Divisor | Quotient | Remainder | Divide Multi-Digit Numbers | divide | multiply | subtract | bring down | repeat | quotient

WATCH OUT!

A common mistake

A common mistake in Divide Multi-Digit Numbers is skipping a placeholder zero in the quotient when a digit doesn't divide evenly, which shifts every digit after it out of place. For example, dividing 3,216 ÷ 3: 3 ÷ 3 = 1, but the next digit 2 doesn't divide by 3, so a 0 must be written before continuing

LESSON 2.6

Divide Multi-Digit Numbers Using an Algorithm

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

WHY THIS MATTERS The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation. How many bars does each section receive?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Run the long-division cycle. Fill in the quotient.

Problem	Divide	Multiply & Subtract	Bring Down	Quotient
384 ÷ 8				
672 ÷ 6				

HINT Read the row left to right — it is one trip around the cycle: divide, multiply, subtract, bring down.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 936 ÷ 12?

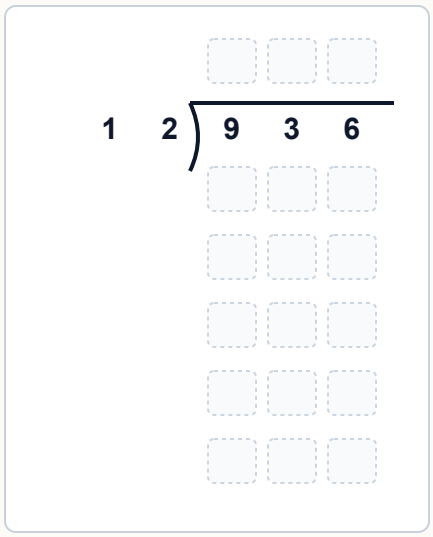
(A) 68

(B) 76

(C) 78

(D) 87

HINT Estimate first: 12 × 70 = 840 and 12 × 80 = 960, so the quotient must land between 70 and 80.



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

3 **Circle the letter of the best answer. Show how you know.**

What is 2,485 ÷ 5?

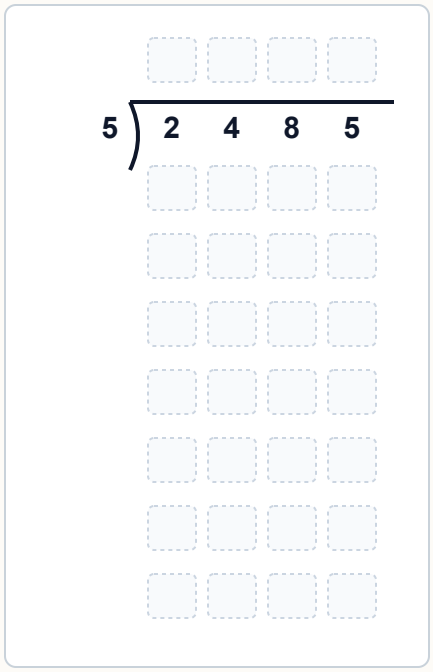
(A) 487

(B) 496

(C) 497

(D) 507

HINT Size it up: 5 × 500 = 2,500, just a bit more than 2,485 — so the quotient is a little under 500.



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

4 **Circle the letter of the best answer. Show how you know.**

What is 756 ÷ 9?

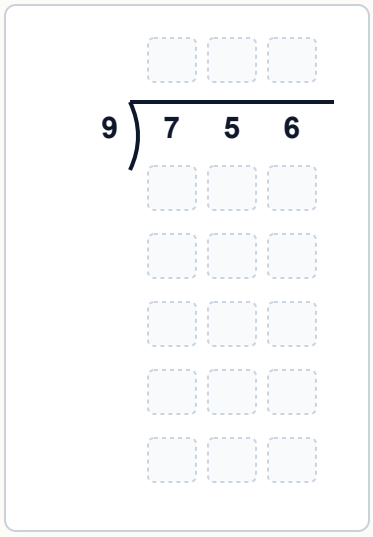
(A) 83

(B) 84

(C) 86

(D) 94

HINT Anchor on a fact you know: 9 × 80 = 720, which is close to 756 — so the quotient is just above 80.



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

5 **Circle the letter of the best answer. Show how you know.**

What is 1,125 ÷ 9?

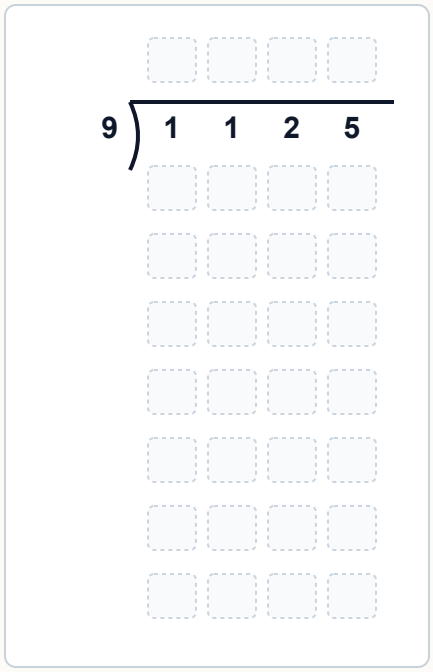
(A) 115

(B) 124

(C) 125

(D) 135

HINT Estimate first: 9 × 100 = 900 and 9 × 200 = 1,800, so the quotient lands between 100 and 200.



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Mistake in 1,134 ÷ 9

1. **Problem** Divide 1,134 ÷ 9 with long division. Dividend = 1,134, divisor = 9.
2. **Divide, Multiply, Subtract** 9 does not fit into 1, so start with 11. 11 ÷ 9 → 1. 1 × 9 = 9, and 11 − 9 = 2
3. **Bring down, repeat** Bring down the 3 to make 23. 23 ÷ 9 → 2. 2 × 9 = 18, and 23 − 18 = 5
4. **Final answer** Quotient = 12, remainder = 5

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Count the digits of 1,134 that have been used so far. Is one of them still waiting to be brought down?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

You ran the cycle three times. What is 2,184 ÷ 14, and how did you know when to stop repeating divide, multiply, subtract, bring down?

SENTENCE STARTER 2,184 ÷ 14 = ___ because I repeated divide, multiply, subtract, and bring down until there were no digits left to bring down, and what was left at the end was ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.6

Divide Multi-Digit Numbers Using an Algorithm

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

WHY THIS MATTERS The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation. How many bars does each section receive?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete each division problem.

Division Problem	Quotient
672 ÷ 8	
2,208 ÷ 16	
2,016 ÷ 14	
1,560 ÷ 15	
3,276 ÷ 12	

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each division problem to its quotient.

_____ 450 ÷ 18
_____ 1,008 ÷ 16
_____ 1,440 ÷ 24
_____ 2,100 ÷ 15
_____ 864 ÷ 12
_____ 1,825 ÷ 25

(A) 25
(B) 63
(C) 60
(D) 140
(E) 72
(F) 73

3 **Complete the table. Show how you found each value.**

Divide each problem with the algorithm, then record the quotient and the remainder.

Division	Quotient	Remainder
144 ÷ 12		
480 ÷ 16		
1,000 ÷ 8		
100 ÷ 7		
500 ÷ 13		
250 ÷ 9		

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A warehouse has 2,184 items to pack into boxes of 14. How many boxes are needed?

(A) 146 boxes

(B) 155 boxes

(C) 156 boxes

(D) 166 boxes

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Mistake in 2,058 ÷ 14

1. **Problem** Divide 2,058 ÷ 14 with long division. Dividend = 2,058, divisor = 14.
2. **Divide, Multiply, Subtract** 14 does not fit into 2, so start with 20. 20 ÷ 14 → 1. 1 × 14 = 14, and 20 − 14 = 6
3. **Bring down, repeat** Bring down the 5 to make 65. 65 ÷ 14 → 4. 4 × 14 = 56, and 65 − 56 = 9
4. **Bring down, repeat** Bring down the 8 to make 98. 98 ÷ 14 → 8. 8 × 14 = 112, and 98 − 112 = −14
5. **Final answer** Quotient = 148

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

You ran the cycle three times. What is 2,184 ÷ 14, and how did you know when to stop repeating divide, multiply, subtract, bring down?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.6

Divide Multi-Digit Numbers Using an Algorithm

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

WHY THIS MATTERS The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation. How many bars does each section receive?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Solve each problem using long division. Show the quotient and any remainder.

Division Problem	Quotient	Remainder
2,578 ÷ 13		
3,456 ÷ 16		
1,987 ÷ 11		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Sonia's Mistake

1. **Problem** 1,536 ÷ 16
2. **Divide** 16 does not fit into 1 or 15, so start with 153. 153 ÷ 16 → 9, written above the 3
3. **Multiply** 9 × 16 = 144, written under the 153
4. **Subtract** 153 − 144 = 19
5. **Bring down** Bring down the 6 to make 196
6. **Final answer** Quotient = 9 with 196 left over

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

The station must load 3,360 ration packs equally into 14 storage pods. Solve it with long division. Name each step — divide, multiply, subtract, bring down — and check your answer by multiplying.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

You ran the cycle three times. What is 2,184 ÷ 14, and how did you know when to stop repeating divide, multiply, subtract, bring down?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help